Code No: 117CF **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, December - 2019
DESIGN PATTERNS
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Write brief notes on usage of design patterns. [2]
b) Discuss how to select a suitable design pattern. [3]
c) What is the applicability of builder patterns? [2]
d) Discuss the structure of singleton pattern. [3]
e) Write the structure of adapter pattern. [2]
f) Write the sample code for bridge pattern. [3]
g) Discuss the consequences of observer pattern. [2]
h) Write implementation of Iterator pattern. [3]
i) Write sample code for template method pattern. [2]
j) Discuss the structure of visitor pattern. [3]

PART-B

(50 Marks)

2. Discuss about design patterns in Smalltalk MVC. [10]

OR

3. What is a design pattern? Discuss how design patterns solve design problems. [10]

4. What are the seven problems in Lexi's user interface design? Explain them clearly.[10]

OR

5. Describe the intent, motivation, structure and applicability of abstract factory pattern.[10]

6. Explain the Intent, participants, consequences and implementation of decorator pattern. [10]

OR

7. Discuss about structure, implementation sample code and known user of flyweight pattern. [10]

8. Write the sample code; known user and related patterns for the claim of responsibility and mediator patterns. [10]

OR

9.a) Discuss in detail about command pattern.
b) Write about the sample code for interpreter pattern. [5+5]

10. Compare state and strategy patterns and implement strategy pattern. [10]

OR

11.a) Explain the history of design patterns.
b) What to expert from design patterns? [6+4]

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